

Lab Experiments

Course Code/Course Name: DSA0604/Data Handling and Visualization

Set 1: Monthly Sales Data

R code:

```
#####
```

```
# SET 1 - MONTHLY SALES DATA
```

```
# Data Handling and Visualization Lab
```

```
#####
```

```
# Install packages (Run only once)
```

```
# install.packages("ggplot2")
```

```
# install.packages("plotly")
```

```
# install.packages("flexdashboard")
```

```
# install.packages("rmarkdown")
```

```
# Load library
```

```
library(ggplot2)
```

```
#####
```

```
# Create Monthly Sales Dataset
```

```
#####
```

```
sales_data <- data.frame(
```

```
  Month = factor(c("January", "February", "March", "April", "May",
```

```
    "June", "July", "August", "September",
```

```
      "October","November","December"),
      levels=c("January","February","March","April","May",
        "June","July","August","September",
        "October","November","December")),
      Sales = c(15000,18000,22000,20000,23000,
        25000,27000,26000,28000,
        30000,32000,35000)
    )
```

```
print(sales_data)
```

```
#####
```

```
# QUESTION 1
```

```
# Line Chart
```

```
#####
```

```
ggplot(sales_data,
  aes(x=Month,y=Sales,group=1))+
  geom_line(color="blue",linewidth=1.2)+
  geom_point(color="red",size=3)+
  labs(title="Monthly Sales",
    x="Month",
    y="Sales (in $)")+
  theme_minimal()
```

```
#####
```

QUESTION 2

Bar Chart - Top Selling Products

#####

```
product_data <- data.frame(  
  Product=c("Laptop","Phone","Tablet",  
            "Headphones","Smart Watch"),  
  Sales=c(50000,70000,30000,20000,40000)  
)
```

```
ggplot(product_data,  
  aes(x=reorder(Product,Sales),  
      y=Sales,  
      fill=Product))+  
geom_bar(stat="identity")+  
coord_flip()+  
labs(title="Top Selling Products",  
     x="Products",  
     y="Annual Sales ($)")+  
theme_minimal()
```

#####

QUESTION 3

Scatter Plot

#####

```
advertising_budget <- c(5000,6000,6500,6200,7000,  
                        7500,8000,8200,8500,  
                        9000,9500,10000)
```

```
scatter_data <- data.frame(  
  Advertising_Budget=advertising_budget,  
  Sales=sales_data$Sales  
)
```

```
ggplot(scatter_data,  
  aes(x=Advertising_Budget,  
      y=Sales))+  
geom_point(color="darkgreen",size=3)+  
geom_smooth(method="lm",  
  se=FALSE,  
  color="red")+  
labs(title="Advertising Budget vs Monthly Sales",  
  x="Advertising Budget ($)",  
  y="Monthly Sales ($)")+  
theme_minimal()
```

```
#####
```

```
# Insight
```

```
#####
```

```
cat("\n")
```

```
cat("INSIGHT:\n")
cat("The scatter plot shows a positive relationship between\n")
cat("advertising budget and monthly sales.\n")
cat("As advertising budget increases, monthly sales also increase.\n")
```

```
#####
```

```
# QUESTION 4
```

```
# Interactive Dashboard
```

```
#####
```

```
# Save the following code in a file named
```

```
# SalesDashboard.Rmd
```

```
# -----
```

```
# ---
```

```
# title: "Monthly Sales Dashboard"
```

```
# output: flexdashboard::flex_dashboard
```

```
# ---
```

```
#
```

```
# ```{r setup, include=FALSE}
```

```
# library(ggplot2)
```

```
# library(plotly)
```

```
#
```

```
# sales_data <- data.frame(
```

```
# Month=factor(c("January","February","March","April","May",
```

```
# "June","July","August","September",
```

```

# "October","November","December"),
# levels=c("January","February","March","April","May",
# "June","July","August","September",
# "October","November","December")),
# Sales=c(15000,18000,22000,20000,23000,
# 25000,27000,26000,28000,
# 30000,32000,35000)
# )
#
# product_data <- data.frame(
# Product=c("Laptop","Phone","Tablet",
# "Headphones","Smart Watch"),
# Sales=c(50000,70000,30000,20000,40000)
# )
# Column
# -----
# ### Monthly Sales
#
# ```{r}
# p1 <- ggplot(sales_data,
# aes(Month,Sales,group=1))+
# geom_line(color="blue")+
# geom_point(color="red",size=3)
#
# ggplotly(p1)
# ```

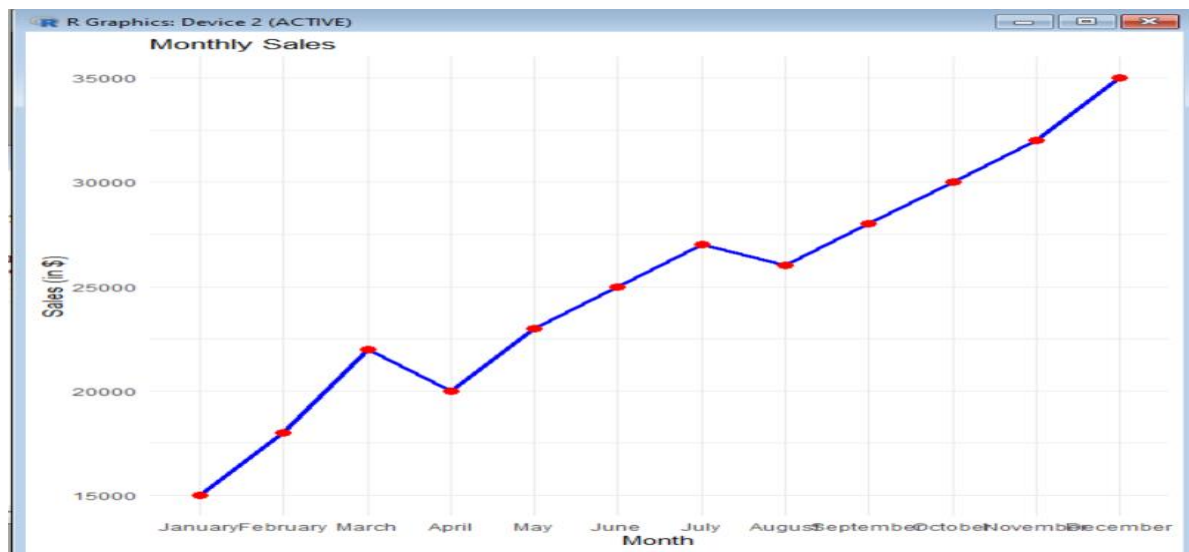
```

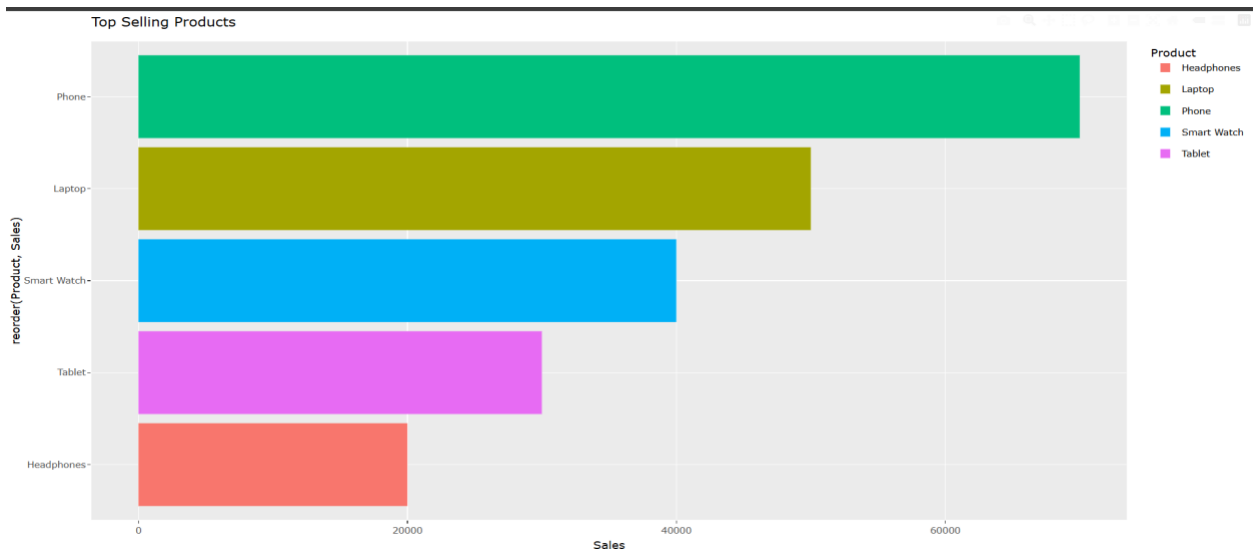
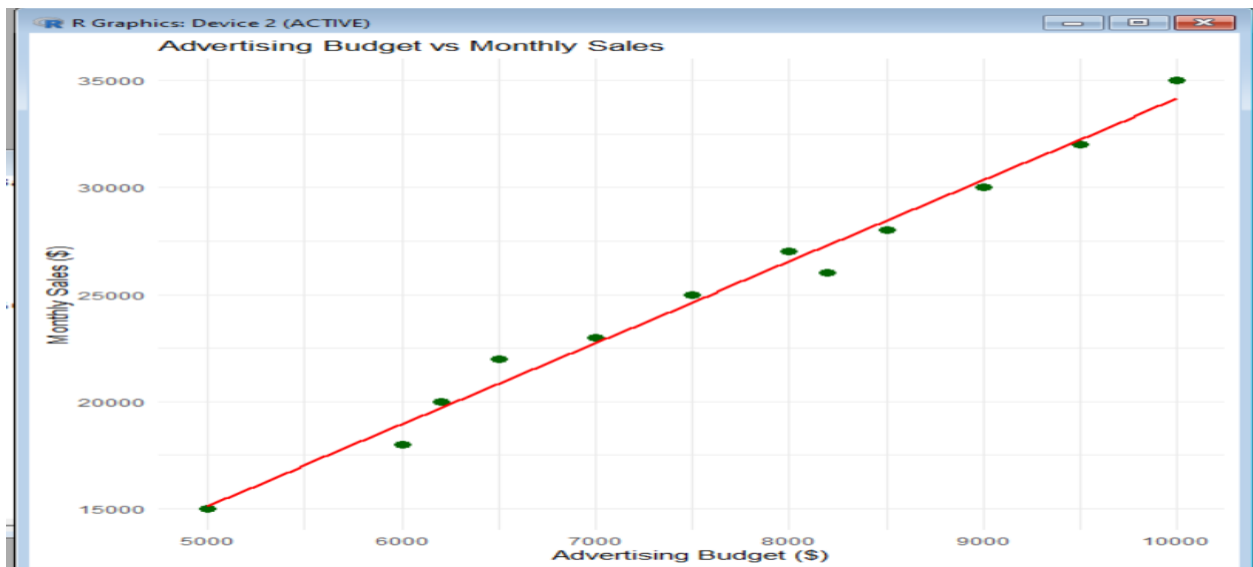
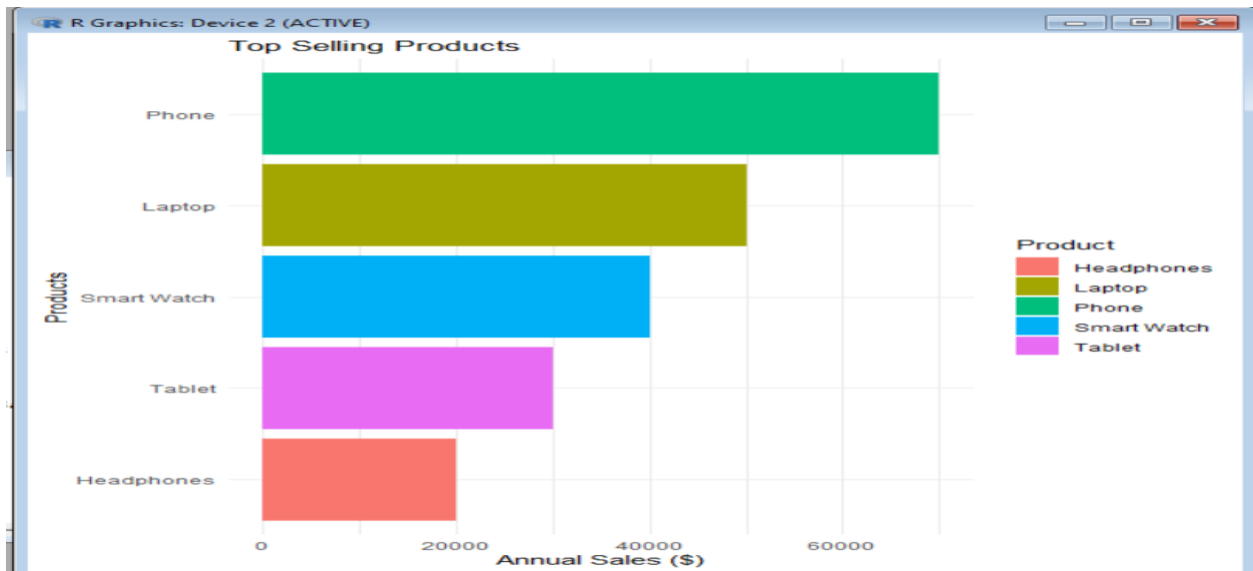
```

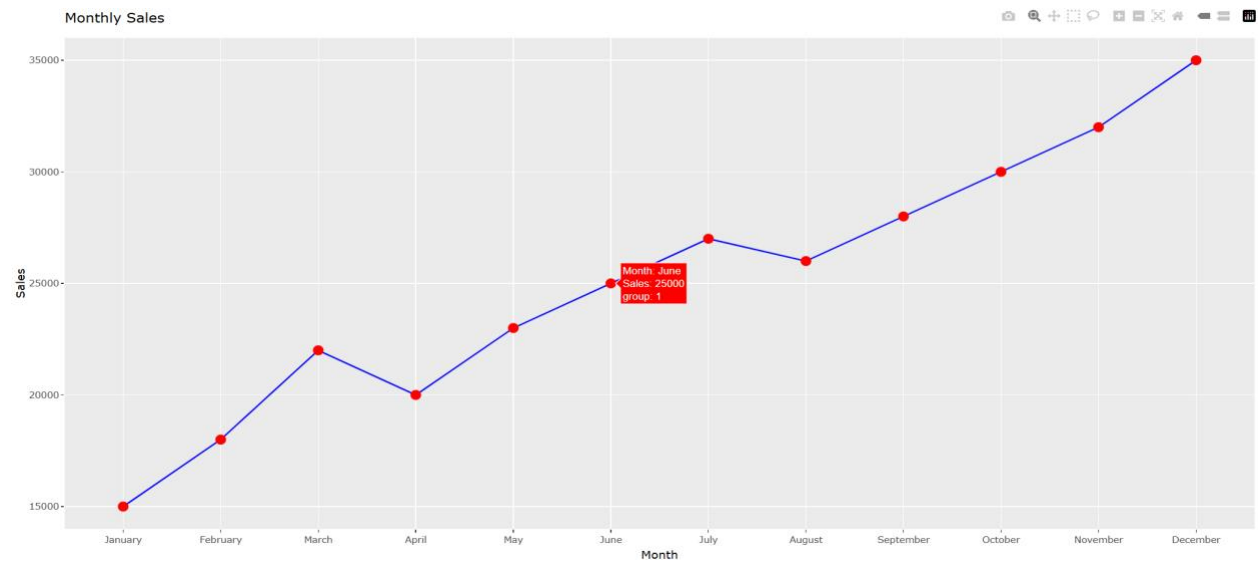
#
# Column
# -----
#
# ### Top Selling Products
#
# ```{r}
# p2 <- ggplot(product_data,
# aes(reorder(Product,Sales),
# Sales,
# fill=Product))+
# geom_bar(stat="identity")+
# coord_flip()
#
# ggplotly(p2)

```

Outputs:







2) Customer Feedback Analysis:

R code:

```
library(ggplot2)
```

```
library(tm)
```

```
library(wordcloud)
```

```
library(RColorBrewer)
```

```
customer_data <- data.frame(
  Customer_ID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction_Score = c(4,5,3,4,5)
)
```

Histogram

```
ggplot(customer_data, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
```

```
labs(  
  title = "Distribution of Customer Ages",  
  x = "Age",  
  y = "Frequency"  
) +  
theme_minimal()
```

Pie Chart

```
score_count <- table(customer_data$Satisfaction_Score)
```

```
pie(  
  score_count,  
  labels = paste("Score", names(score_count)),  
  col = rainbow(length(score_count)),  
  main = "Customer Satisfaction Score Distribution"  
)
```

Stacked Bar Chart

```
customer_data$Age_Group <- cut(  
  customer_data$Age,  
  breaks = c(20,30,40),  
  labels = c("20-30", "31-40"),  
  include.lowest = TRUE  
)
```

```
table_data <- table(
```

```
customer_data$Age_Group,  
customer_data$Satisfaction_Score  
)
```

```
barplot(  
  table_data,  
  col = c("orange","lightblue"),  
  legend.text = rownames(table_data),  
  main = "Customer Satisfaction by Age Group",  
  xlab = "Satisfaction Score",  
  ylab = "Number of Customers"  
)
```

Word Cloud

```
feedback <- c(  
  "Excellent service",  
  "Very satisfied",  
  "Good quality",  
  "Fast delivery",  
  "Friendly staff",  
  "Excellent support",  
  "Affordable price",  
  "Needs improvement",  
  "Great experience",  
  "Satisfied overall"  
)
```

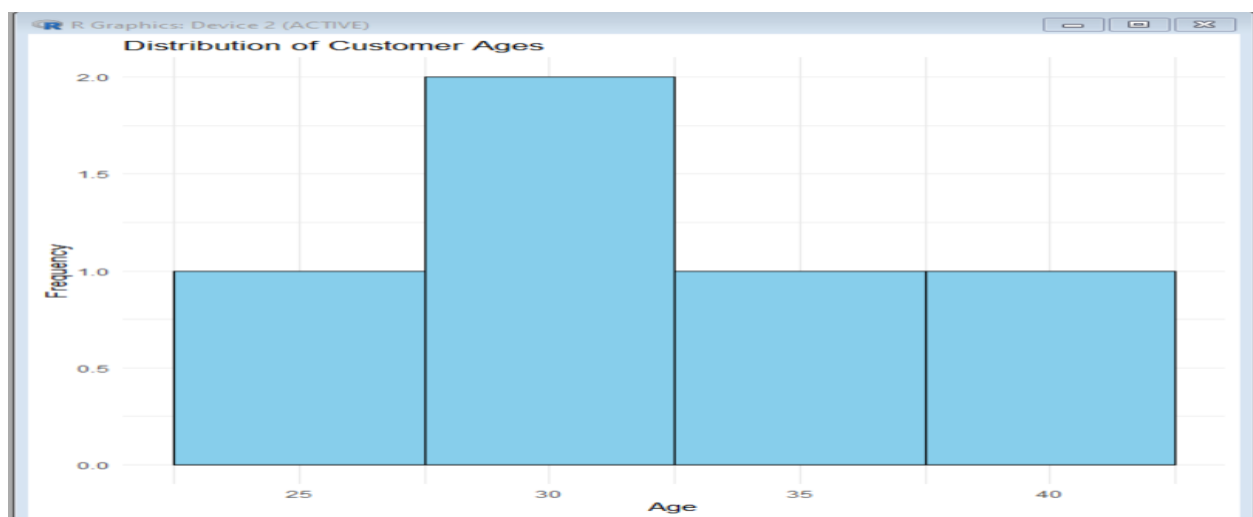
```

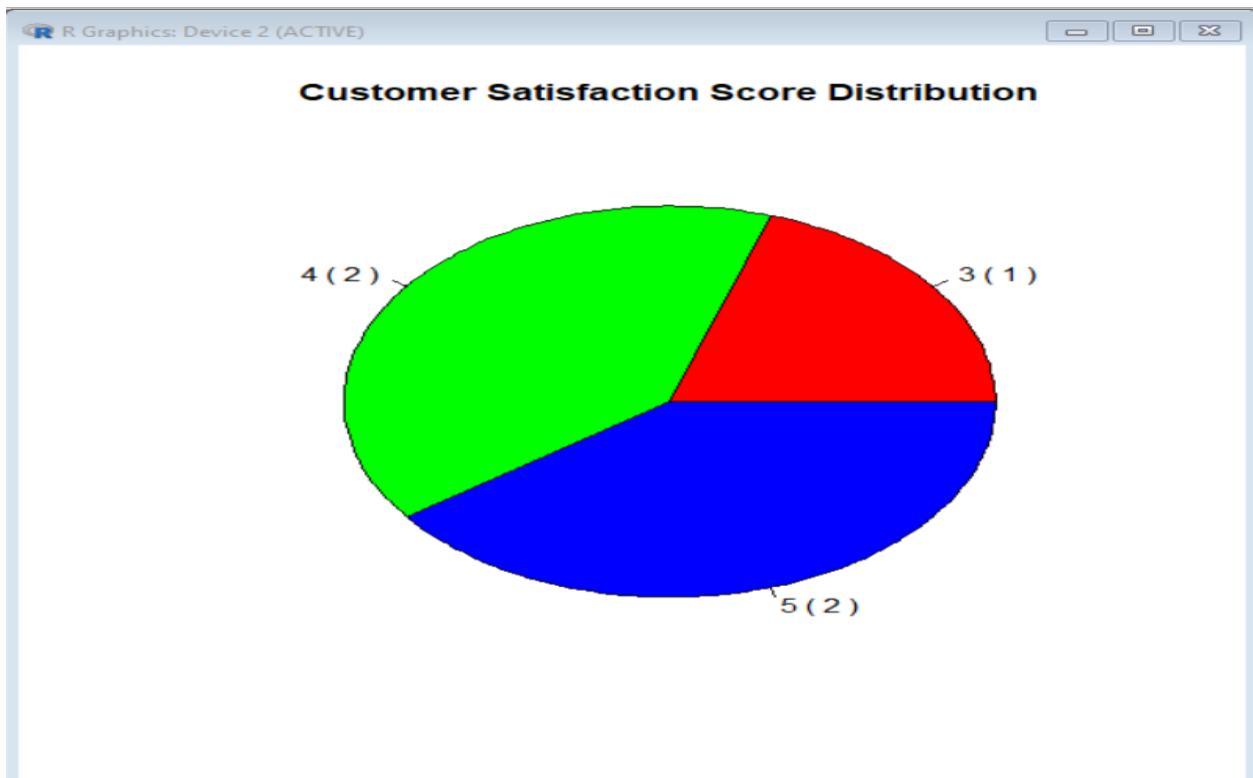
corpus <- Corpus(VectorSource(feedback))
corpus <- tm_map(corpus, content_transformer(tolower))
corpus <- tm_map(corpus, removePunctuation)
corpus <- tm_map(corpus, removeNumbers)
corpus <- tm_map(corpus, removeWords, stopwords("english"))
tdm <- TermDocumentMatrix(corpus)
m <- as.matrix(tdm)
word_freq <- sort(rowSums(m), decreasing = TRUE)

wordcloud(
  words = names(word_freq),
  freq = word_freq,
  min.freq = 1,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2")
)

```

Output:







Set 3: Employee Performance Evaluation

R code:

```
library(ggplot2)
```

```
employee_data <- data.frame(  
  Employee_ID = c(1,2,3,4,5),  
  Department = c("Sales","HR","Marketing","Sales","HR"),  
  Years_of_Service = c(5,3,7,4,2),  
  Performance_Score = c(85,92,78,90,76)  
)
```

Line Chart

```
ggplot(employee_data,  
  aes(x = Employee_ID,  
    y = Performance_Score,
```

```
    group = 1)) +  
geom_line(color = "blue", linewidth = 1.2) +  
geom_point(color = "red", size = 3) +  
labs(  
  title = "Employee Performance Trend",  
  x = "Employee ID",  
  y = "Performance Score"  
) +  
theme_minimal()
```

Bar Chart

```
ggplot(employee_data,  
  aes(x = Department,  
    fill = Department)) +  
geom_bar() +  
labs(  
  title = "Employees Across Departments",  
  x = "Department",  
  y = "Number of Employees"  
) +  
theme_minimal()
```

Scatter Plot

```
ggplot(employee_data,  
  aes(x = Years_of_Service,  
    y = Performance_Score)) +
```

```

geom_point(color = "darkgreen", size = 4) +
geom_smooth(method = "lm",
             se = FALSE,
             color = "red") +
labs(
  title = "Years of Service vs Performance Score",
  x = "Years of Service",
  y = "Performance Score"
) +
theme_minimal()
# Install packages (Run once)
# install.packages("flexdashboard")
# install.packages("plotly")

library(ggplot2)
library(plotly)

employee_data <- data.frame(
  Employee_ID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service = c(5,3,7,4,2),
  Performance_Score = c(85,92,78,90,76)
)

# Line Chart
p1 <- ggplot(employee_data,

```



```
aes(x = Employee_ID,  
    y = Performance_Score,  
    group = 1)) +  
geom_line(color = "blue") +  
geom_point(color = "red", size = 3) +  
labs(title = "Employee Performance Trend")
```

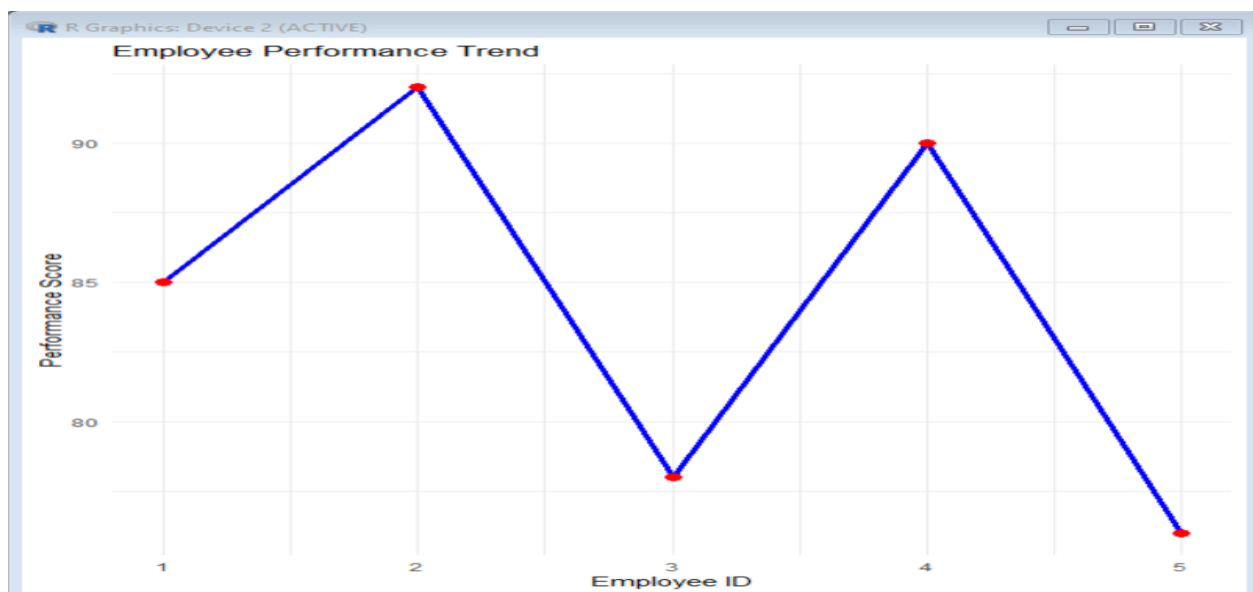
```
ggplotly(p1)
```

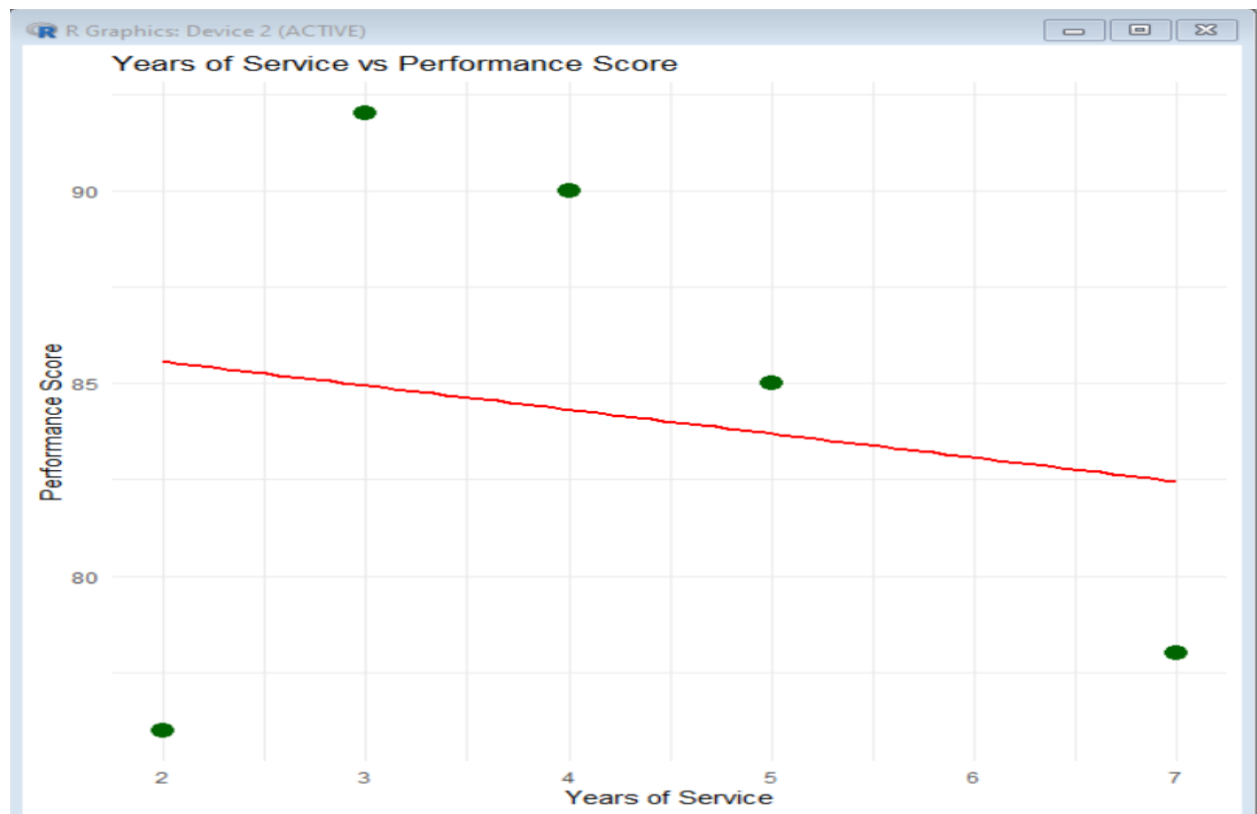
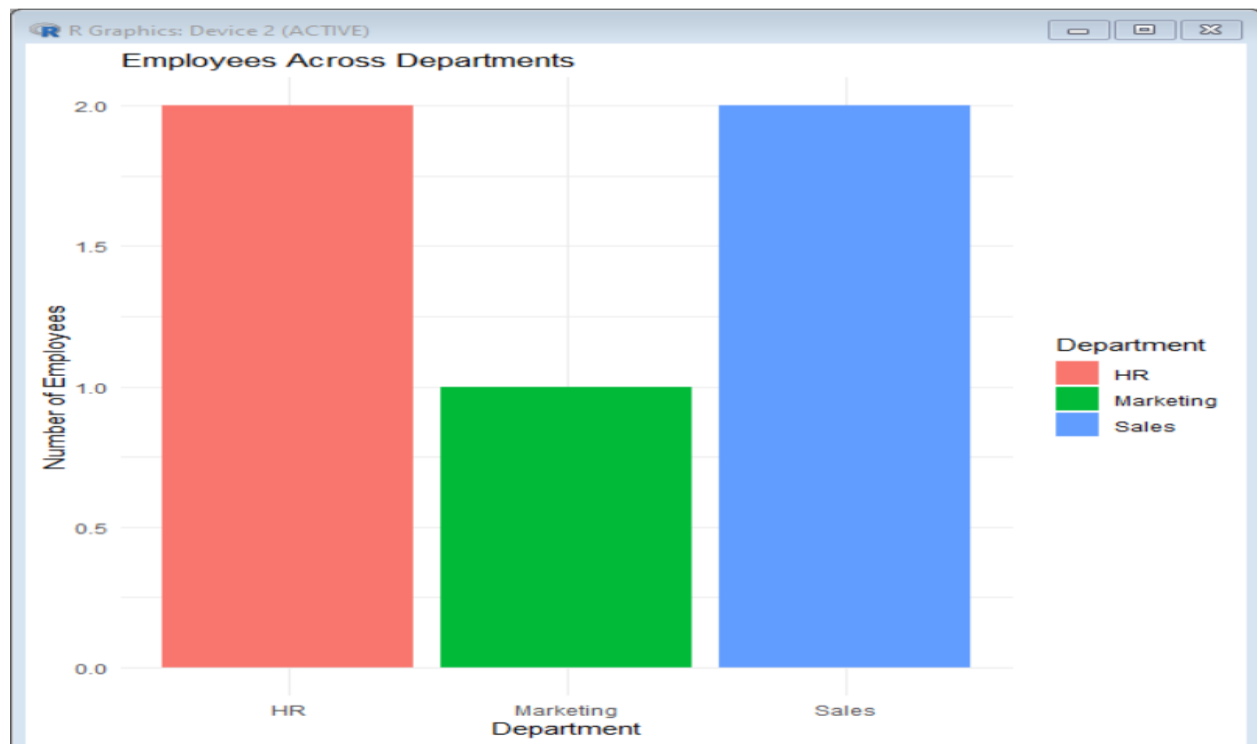
Bar Chart

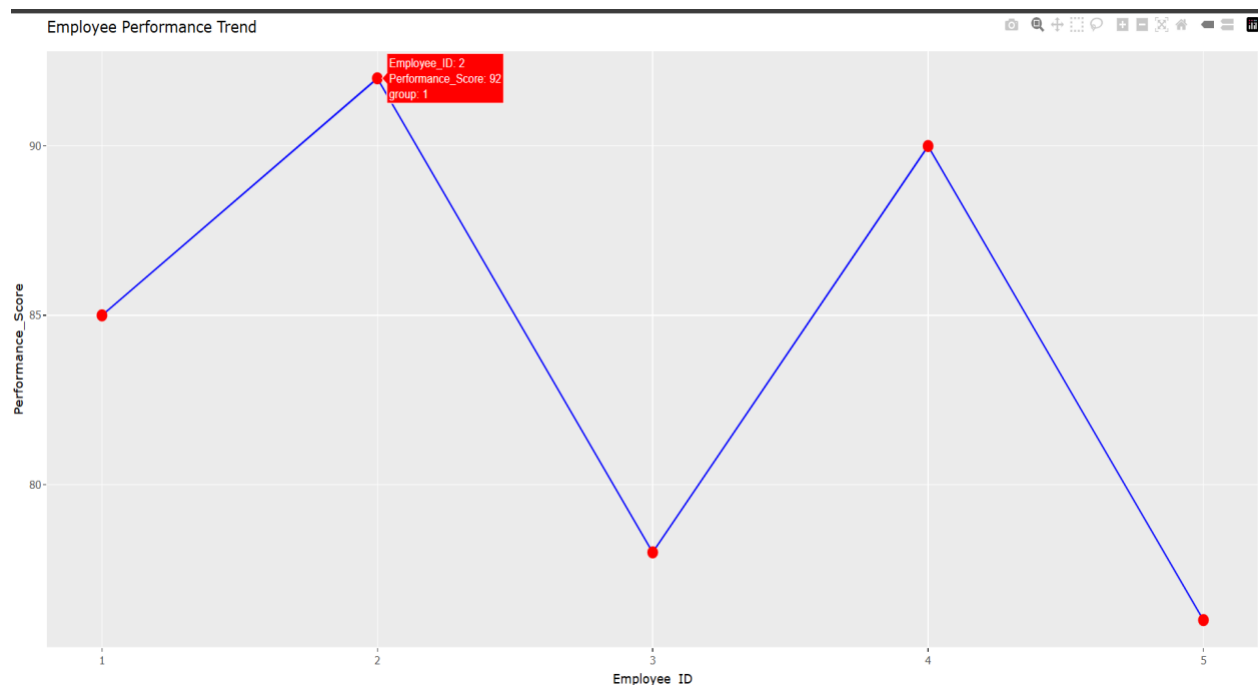
```
p2 <- ggplot(employee_data,  
             aes(x = Department,  
                 fill = Department)) +  
geom_bar() +  
labs(title = "Employees Across Departments")
```

```
ggplotly(p2)
```

Output:







Set 4: Product Inventory Management

R code:

```
library(ggplot2)
```

```
product_data <- data.frame(
```

```
  Product_ID = c(1,2,3,4,5),
```

```
  Product_Name = c("Product A","Product B","Product C","Product D","Product E"),
```

```
  Category = c("Category 1","Category 1","Category 2","Category 2","Category 3"),
```

```
  Quantity = c(250,175,300,200,220),
```

```
  Price = c(20,15,18,22,19)
```

```
)
```

```
# Bar Chart
```

```
ggplot(product_data,
```

```
  aes(x = Product_Name,
```

```
    y = Quantity,
```

```
    fill = Product_Name)) +  
geom_bar(stat = "identity") +  
labs(  
  title = "Product Inventory Quantity",  
  x = "Product",  
  y = "Quantity Available"  
) +  
theme_minimal()
```

Stacked Bar Chart

```
ggplot(product_data,  
  aes(x = Category,  
    y = Quantity,  
    fill = Product_Name)) +  
geom_bar(stat = "identity") +  
labs(  
  title = "Quantity by Product Category",  
  x = "Category",  
  y = "Quantity"  
) +  
theme_minimal()
```

Scatter Plot

```
ggplot(product_data,  
  aes(x = Price,  
    y = Quantity)) +
```

```
geom_point(color = "blue", size = 4) +  
geom_smooth(method = "lm",  
            se = FALSE,  
            color = "red") +
```

```
labs(  
  title = "Price vs Quantity Available",  
  x = "Price ($)",  
  y = "Quantity Available"  
) +  
theme_minimal()
```

```
# Install packages (Run once)
```

```
# install.packages("flexdashboard")
```

```
# install.packages("plotly")
```

```
library(ggplot2)
```

```
library(plotly)
```

```
product_data <- data.frame(  
  Product_ID = c(1,2,3,4,5),  
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Category = c("Category 1", "Category 1", "Category 2", "Category 2", "Category 3"),  
  Quantity = c(250,175,300,200,220),  
  Price = c(20,15,18,22,19)  
)
```

```
# Bar Chart
```

```
p1 <- ggplot(product_data,
  aes(x = Product_Name,
    y = Quantity,
    fill = Product_Name)) +
  geom_bar(stat = "identity") +
  labs(title = "Product Inventory Quantity")
```

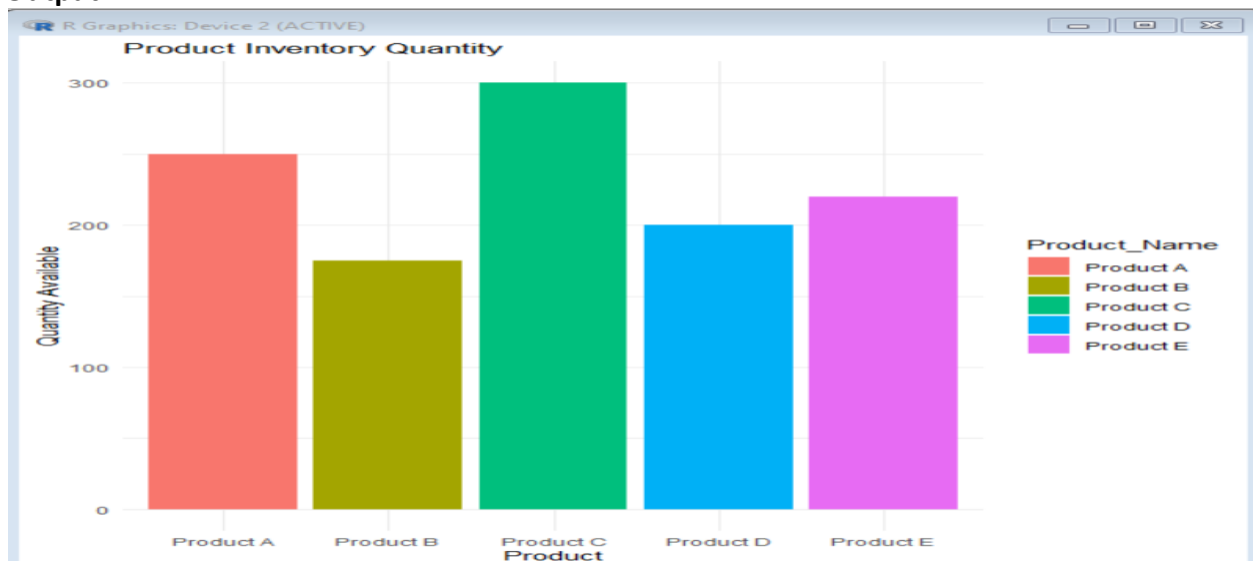
```
ggplotly(p1)
```

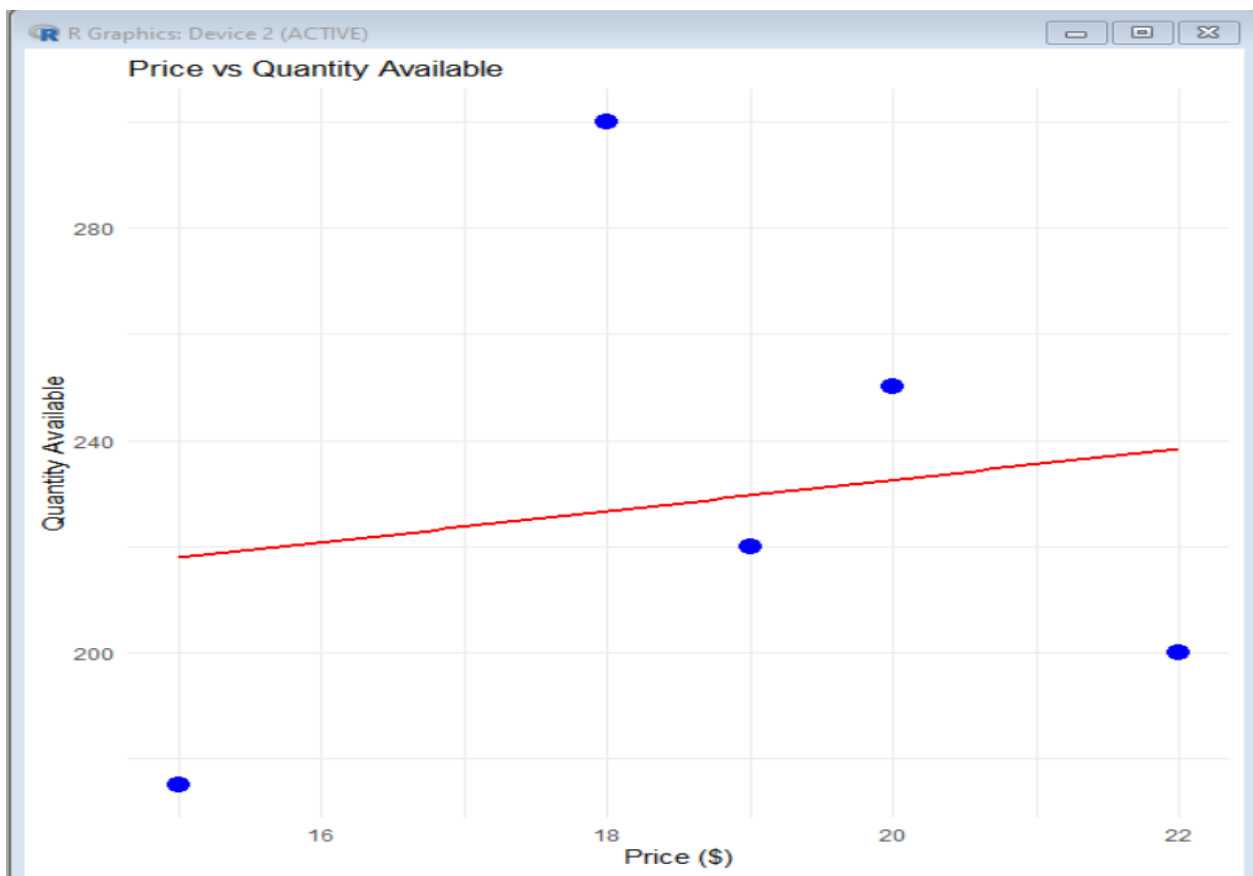
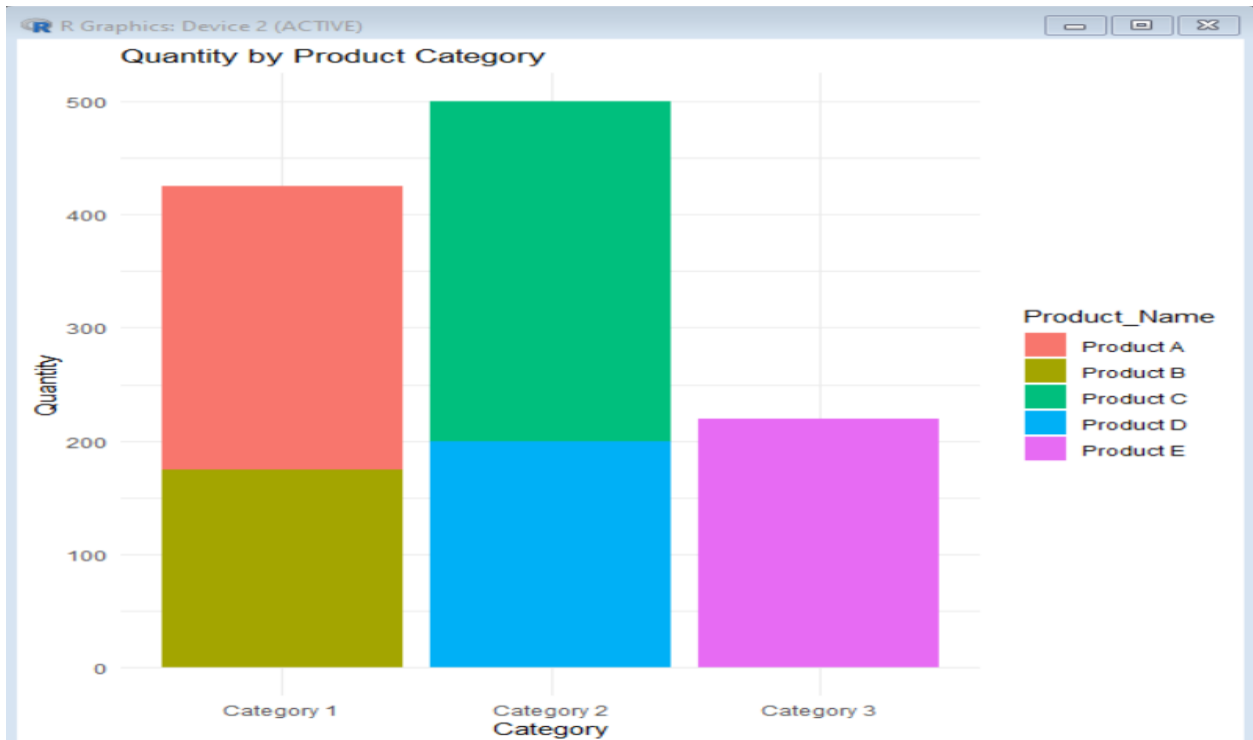
Stacked Bar Chart

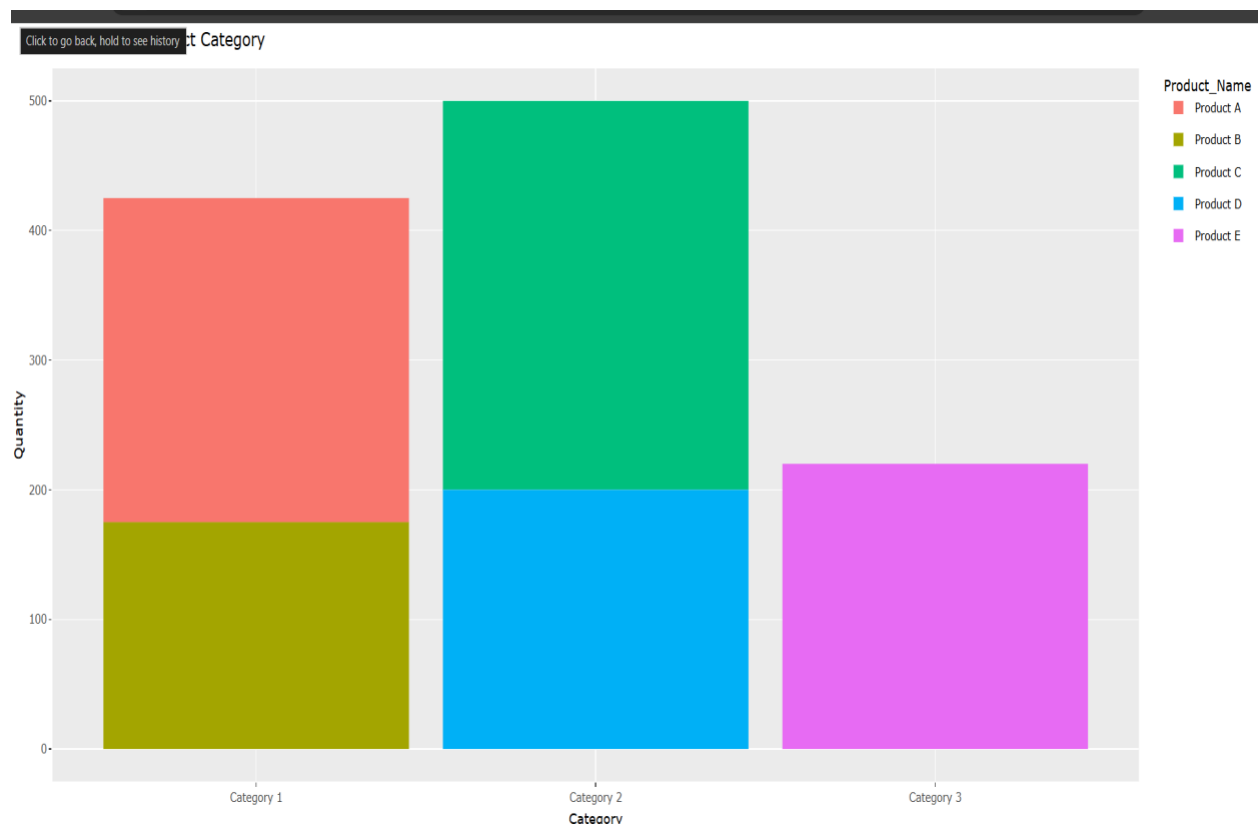
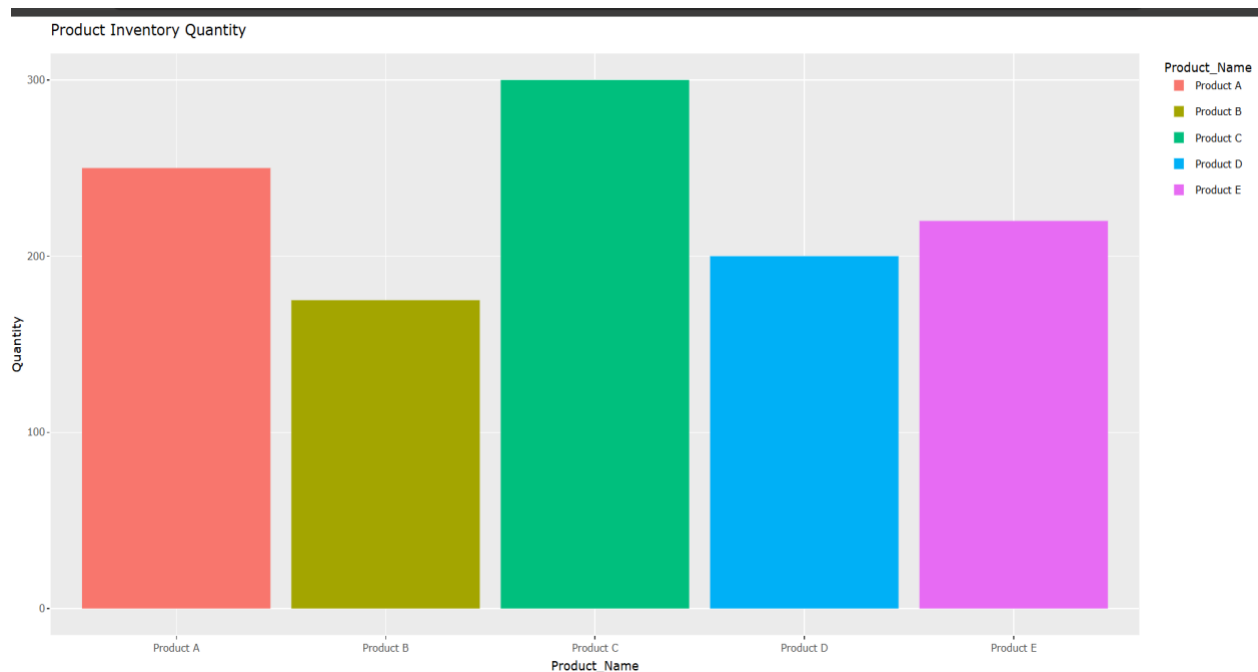
```
p2 <- ggplot(product_data,
  aes(x = Category,
    y = Quantity,
    fill = Product_Name)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity by Product Category")
```

```
ggplotly(p2)
```

Output:







Set 5: Website Analytics

R code:

```
library(ggplot2)
```

```
website_data <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),  
  Page_Views = c(1500, 1600, 1400, 1650, 1800),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6),  
  Likes = c(120, 140, 110, 150, 170),  
  Shares = c(60, 75, 50, 80, 90),  
  Comments = c(30, 35, 25, 40, 45)  
)
```

Line Chart

```
ggplot(website_data, aes(x = Date, y = Page_Views, group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Daily Page Views",  
    x = "Date",  
    y = "Page Views"  
  ) +  
  theme_minimal()
```

Bar Chart (Top Click-Through Rates)

```
ggplot(website_data,
```

```

aes(x = reorder(as.character(Date), CTR),
    y = CTR,
    fill = as.character(Date))) +
geom_bar(stat = "identity") +
labs(
  title = "Top Click-Through Rates",
  x = "Date",
  y = "Click-Through Rate (%)"
) +
theme_minimal()

```

Stacked Area Chart

```
library(tidyr)
```

```

interaction_data <- pivot_longer(
  website_data,
  cols = c(Likes, Shares, Comments),
  names_to = "Interaction",
  values_to = "Count"
)

```

```

ggplot(interaction_data,
  aes(x = Date,
      y = Count,
      fill = Interaction)) +
geom_area() +

```

```

labs(
  title = "Website User Interactions",
  x = "Date",
  y = "Count"
) +
  theme_minimal()

# Install packages (Run once)
# install.packages("flexdashboard")
# install.packages("plotly")

library(ggplot2)
library(plotly)

website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

# Line Chart
p1 <- ggplot(website_data,
  aes(x = Date,
    y = Page_Views,
    group = 1)) +
  geom_line(color = "blue") +
  geom_point(color = "red", size = 3) +

```

```
labs(title = "Daily Page Views")
```

```
ggplotly(p1)
```

```
# Bar Chart
```

```
p2 <- ggplot(website_data,  
  aes(x = reorder(as.character(Date), CTR),  
    y = CTR,  
    fill = as.character(Date))) +  
  geom_bar(stat = "identity") +  
  labs(title = "Top Click-Through Rates")
```

```
ggplotly(p2)
```

Output:

